

Team Members: 02 (only Team Lead submit assignment)

Deadline: 18 Sep 2026

Objective:

Students will apply **Software Construction** concepts by building a **small but complete software system from scratch**.

ExpenseMate — Personal Expense Manager

Idea: Income/expense record kare, monthly budgets banaye, aur visual charts dikhaye.

Aspect	Detail
Architecture	Client-Server (local)
Tech	Python + SQLite / C# + SQL etc.
Challenging part	Budget alerts, CSV import/export, category-wise analytics
Maintenance scope	Add multi-currency, cloud backup (Optional)

Requirements → Design → Construction → Testing → Maintenance

Assignment Breakdown

Phase 1: Problem Definition & Requirements

Task	Who
Write functional & non-functional requirements	Member A
Create use case diagram	Member B

Deliverable: 2-page SRS (Software Requirements Specification)

Phase 2: Architecture & Construction Planning

Task	Who
Choose an architectural style (Layered / Client-Server / Peer-to-Peer)	Both
Draw Logical, Process, Physical, Deployment views	Member A
Plan coding standards, tools, and timeline	Member B

Deliverable: Architecture document + Gantt chart

Phase 3: Coding & Unit Testing

Task	Who
Divide system into modules (e.g., UI, Logic, DB)	Both
Write code (Python/Java/C++)	Member A: Backend, Member B: Frontend
Write unit tests for each module	Each tests their own

Deliverable: Source code + Unit test reports

Phase 4: Integration & System Testing

Task	Who
Integrate modules	Both
Perform integration testing	Member A
Perform system testing	Member B
Measure code complexity (e.g., cyclomatic complexity)	Member A
Track fault-find & fault-fix rates	Member B

Deliverable: Test cases + Bug report + Complexity metrics

Phase 5: Maintenance & Evolution

Task	Who
Add one new feature (adaptive maintenance)	Both
Fix 2 bugs (corrective maintenance)	Member A
Improve code quality (perfective maintenance)	Member B
Apply Lehman's Laws analysis	Both

Deliverable: Change log + Refactoring report

Phase 6: Final Presentation & Report

Task	Who
Write final report (03 pages)	Member B
Demo the working system	Both

Final Submission Bundle

- SRS Document
- Architecture Diagrams
- Source Code (ZIP)
- Test Cases + Results
- Metrics Report
- Final Report